PORTFOLIO CASE STUDY: PROJECT 2

AI-Powered Customer Support Agent & CRM Automated Booking Bot

Executive Summary

This project delivers an end-to-end autonomous customer onboarding system. It combines an intelligent conversational AI layer with multi-system automation, securely linking frontend chat interaction straight to an enterprise CRM and team scheduling infrastructure.

Core Architecture & Technology Stack

Component / Layer	Technology Used	Operational Role
Frontend / AI Agent	Voiceflow (Agentic Core v4.2)	Handles 24/7 customer query support via a dynamic Knowledge Base (RAG context). Captures customer name and email dynamically.
Middleware Automation	Make.com (Webhooks Engine)	Acts as the secure automation bridge that catches payload data, processes maps, and drives external system nodes.
CRM Infrastructure	HubSpot CRM	Instantly receives user payloads via Make.com and updates/creates customer contact records in real-time.
Scheduling System	Calendly Platform	Provides dynamic appointment setting with dynamic parameter injection for user pre-fill to increase conversion.

System Execution Flow & Logic

- **Step 1 - AI Greetings & Intent Selection:** The bot welcomes the user via Voiceflow, offering options to answer queries or directly arrange a strategic booking session.
- **Step 2 - Autonomous Harvesting:** If booking is chosen, the agent runs a conversational playbook extracting the user's Full Name and Verified Email Address.
- **Step 3 - Secure API Relay:** Data is passed as a structural JSON payload via a `POST` request triggered natively by Voiceflow to a custom Make.com Webhook listener.
- **Step 4 - Contact Injection:** Make.com triggers the HubSpot CRM node, verifying and creating the Contact Profile instantaneously in the background repository.
- **Step 5 - Pre-filled URLs Handshake:** The integration yields a structured Webhook Response returning a custom tracking Calendly link with pre-mapped query parameters back to Voiceflow, which is presented as a high-conversion clickable card to the client.

Business Value & Results

Implementing this automated infrastructure saves up to 80% of human onboarding overhead, securely mitigates manual operational friction, and maximizes sales pipeline velocity by delivering an optimal booking framework 24/7.